NOLAN POZZOBON

EDUCATION

Carnegie Mellon University, Pittsburgh, PA *Expected May 2028*
M.S. Artificial Intelligence and Innovation (MSAII)

The University of Chicago, Chicago, IL *June 2026*
B.S. Computer Science (Machine Learning specialization); B.A. Linguistics — GPA: 3.81 / 4.00
Relevant Coursework: Algorithms, Discrete Mathematics, Distributed Systems, Database Systems, Machine Learning, Natural Language Processing, Math for Machine Learning, Large Language Models, Adversarial AI, Computational Linguistics
Awards: Scholar All-American, NCAA Wrestling (2024, 2026)

TECHNICAL SKILLS

Languages: Python, Rust, SQL, JavaScript
ML / AI: PyTorch, Hugging Face Transformers, vLLM, LoRA/PEFT, RAG, LangGraph/LangChain, LLM evaluation
Tools & Infrastructure: Git, Docker, Linux, AWS Lambda, Modal, Lambda Labs, PostgreSQL, Neo4j, Vue, NumPy, scikit-learn

EXPERIENCE

Opteryx — AI Engineering Fellow *June 2026 – August 2026*
Valencia, Spain (Hybrid)
- Shipped a document-humanization feature into a production LLM pipeline generating regulated pharmaceutical stability reports for paying enterprise customers.
- Built feature extraction over heterogeneous document corpora to model stylistic and structural patterns, and rearchitected the pipeline to consume it.

UPS — Supply Chain Solutions Intern *July 2025 – September 2025*
Alpharetta, GA
- Applied mixed-integer linear programming and clustering algorithms to franchise region assignment across the US, reducing operating cost by $100,000 annually.
- Implemented a constrained KNN algorithm to group packages by ZIP code and size range, improving last-mile delivery routing efficiency nationwide.

Toyota Technological Institute at Chicago — Research Assistant *January 2025 – June 2025*
Chicago, IL — Advisor: Dr. Zhewei Sun
- Extended LLM architectures in PyTorch across distributed GPUs on the TTIC cluster, generalizing slang-embedding research to a multilingual setting.

CharacTour — Software Engineering Intern *June 2024 – August 2024*
- Built generative-AI tooling to rewrite platform content for 300 distinct personality profiles, and refined the production recommendation matching algorithm.

ESGX — Product Engineering Intern *August 2024 – January 2025*

RESEARCH

“On Cross-Lingual Regularities in Slang” — ACL Rolling Review *Under review, 2026*
with Dr. Zhewei Sun, Toyota Technological Institute at Chicago
- Assembled novel multilingual slang datasets through custom web-scraping and cleaning pipelines used for model fine-tuning and evaluation.

PROJECTS

Multilingual Sycophancy and Commitment Failure (PyTorch) — sole author *March 2026 – May 2026*
- Evaluated 11 models across 7 languages on capitulation under user pressure, localizing the mechanism with linear probes, the logit lens, and cross-language activation patching.
- Found capitulation scales inversely with language resource level while the correct answer remains recoverable from mid-layer activations — a commitment failure rather than a retrieval failure.

Nabu — AI Language Tutor (Python, LangGraph, Whisper, ElevenLabs) *July 2025 – September 2025*
- Built an AI-native desktop app for real-time spoken conversation, designing the agent workflows, prompt orchestration, and API endpoints across three speech and language services.

Multi-Agent Claim Evaluation System (Python, RAG, NLI) *September 2025 – December 2025*
- Engineered a multi-agent pipeline combining natural language inference, retrieval over embedded user histories, and external grounding to flag unsupported claims.
- Benchmarked across Gemma 3, Llama 3.1, and GPT-OSS with ablations and a human-annotated evaluation set, isolating retrieval quality as the primary bottleneck.

Targeted Knowledge Updates in LLMs (PyTorch, LoRA) *September 2025 – December 2025*
- Implemented inference-time control over the edit–preservation tradeoff via an alpha sweep on LoRA task vectors: 100% edit efficacy at 87% broad task accuracy, no retraining.
- Validated capability isolation with a 50-item multi-domain locality set, preserving 100% code accuracy and near-baseline perplexity.